Hospital Operations & Patient Analysis Dashboard

Project Overview

This Power BI dashboard provides a comprehensive 360-degree view of hospital operations, financial health, and patient demographics. It is designed to help healthcare administrators identify operational bottlenecks, optimize staffing, and improve revenue cycle management.

Data Transformation & Engineering

Before building the visuals, extensive data cleaning and standardization were performed in **Power Query** and **DAX** to ensure data integrity and usability.

1. Data Cleaning & Standardization

- **Privacy & Optimization:** Removed unnecessary Personal Identifiable Information (PII) and redundant columns, including `address`, `phone number`, and `email id` to streamline the model and prioritize data privacy.
- **Name Normalization:** Merged `first_name` and `last_name` columns into a single `Full Name` field for better readability in reports.
- **Value Mapping:** Standardized the `Gender` column by converting abbreviated values ("M", "F") to full descriptive terms ("Male", "Female") for professional presentation.

2. Feature Engineering (DAX)

- **Dynamic Age Calculation:** Created a calculated column to determine patient age using the `date_of_birth` field.
- **Age Grouping:** Categorized patients into meaningful demographic segments using the following logic:

  ```
  AgeGroup=SWITCH(TRUE(),[Age]<=18,"0−18",[Age]<=35,"19−35",[Age]<=50,"36−50",[Age]<=65,"51−65","65+")
  ```

- **Time Intelligence:** Developed a dedicated **Date Table** to enable advanced time-series analysis and ensure accurate monthly/seasonal trending.
- **Shift Categorization:** Grouped raw `appointment_time` data into operational shifts to identify peak bottlenecks:

  ```
  TimePeriod=SWITCH(TRUE(),HOUR([time])≥5 & <12,"Morning",HOUR([time])≥12 & <17,"Afternoon",HOUR([time])≥17 & <21,"Evening","Night")
  ```

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3. Data Modeling

- **Schema:** Implemented a **Star Schema** to optimize query performance.
- **Relationships:** Established one-to-many relationships between Dimension tables (Patients, Doctors, Calendar) and Fact tables (Appointments, Billing).

Key Insights & Business Questions Answered

Patient Demographics

- **Total Registrations:** 50 unique patients are currently registered in the system.
- **Primary Demographic:** The **19-35** demographic is actually your largest patient group, with nearly 20 patients. The **65+** demographic is actually your smallest represented group on this chart.
- **Gender Distribution:** The base consists of **31 Male (62%)** and **19 Female (38%)** patients.
- **Insurance Coverage: MedCare Plus** and **WellnessCorp** are the most common insurance providers.

Appointment Analysis

- **Total Volume:** 200 total appointments have been recorded.
- **Status Breakdown: 23%** of appointments are successfully completed, while **25.5%** are cancelled.
- **No-Show Rate:** A significant **26% no-show rate** highlights a key area for operational improvement.
- **Seasonal Trends: April** experienced a massive volume surge, marking the highest peak in the calendar year. The lowest point in appointment volume is indeed in **September**.
- **Peak Hours:** Daily operations peak during the **Afternoon**, followed by the Morning shift.

Doctor & Staff Performance

- **Top Overall Performer**: **Sarah Taylor** handles the highest total volume of patient appointments across the organization.

- **Branch-Level Leaders**:

  - **Central Hospital**: **Sarah Taylor** is the top-performing physician.
  - **Eastside Clinic**: **Jane Smith** leads in appointment volume.

- **Westside Clinic**: **David Taylor** handles the most patients.
- **Key Specializations: Oncology** is the most frequent specializations.
- **Branch Distribution: Central Hospital** maintains the highest concentration of medical staff.

Treatment & Medical Services

- **High-Frequency Treatment: Chemotherapy** is the most frequently performed procedure.
- **High-Value Services: MRI and Specialized Scans** drive the highest individual costs and profit margins per procedure.

Financial Overview

- **Total Revenue:** The hospital has generated **$551.2K** in total revenue.
- **Billing Efficiency:** The average bill amount per patient stands at **$2.8K**.
- **Revenue Drivers:** Due to high frequency, **Chemotherapy** is the leading contributor to total revenue.
- **Payment Preferences: Credit Cards** are the most used payment method, followed by Insurance.

Technical Features

- **Star Schema Modeling:** Organized data into Fact and Dimension tables for optimized performance.
- **Drill-Through Reporting:** Detailed page navigation to analyze specific doctor and patient records.
- **Time Intelligence:** Created a custom DAX Calendar table to track monthly and seasonal trends.
- **Advanced DAX:** Implemented measures for Total Revenue, Average Billing, and Status Percentages.

Advanced Interactivity: Drill-Through Reporting

A key technical feature of this dashboard is the **Doctor Performance Drill-Through**:

- **Granular Deep-Dive:** Users can right-click on any doctor in the main "Appointments by Doctor" chart to navigate to a dedicated detail page.
- **Specific Metrics:** The drill-through page filters all visuals—including monthly trends and treatment types—to show data specific to that individual physician.
- **Actionable Data:** This allows management to conduct performance reviews by seeing a doctor's specific revenue contribution and patient satisfaction levels.

Conclusion & Recommendations

Based on the data-driven insights from this analysis, the following strategic actions are recommended to optimize hospital performance:

- **Operational Efficiency:** Implement an automated SMS/Email reminder system to target the **23% no-show rate**, which represents a significant leak in potential revenue.
- **Resource Allocation:** Rebalance staffing schedules to address the **Afternoon peak**, where over 50% of daily appointments are concentrated.
- **Market Growth:** Develop specialized wellness or preventative care packages specifically for the **19–35 age demographic**, as they represent the largest portion of the current patient base.
- **Revenue Optimization:** Expand the availability of **Chemotherapy and MRI services** during underutilized Evening slots to maximize the high profit margins associated with these treatments.